

[Chathurya V R 251801045-2025-AIDS](#) ▾**C2****Started on** Thursday, 17 September 2026, 8:22 AM**State** Finished**Completed on** Thursday, 17 September 2026, 8:23 AM**Time taken** 49 secs**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 10.00 out of 10.00**Playing with Numbers:**

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3. Write any efficient algorithm to find the possible ways.

Example 1:**Input:** 6**Output:** 6**Explanation:** There are 6 ways to 6 represent number with 1 and 3

1+1+1+1+1+1

3+3

1+1+1+3

1+1+3+1

1+3+1+1

3+1+1+1

Input Format

First Line contains the number n

Output Format**Print: The number of possible ways 'n' can be represented using 1 and 3**

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5
6      // Read the input silently without any extra text prompts
7      if (scanf("%d", &n) != 1) return 0;
8
9      // Handle small inputs directly using base cases
10     if (n == 0) {
11         printf("1\n");
12         return 0;
13     } else if (n == 1 || n == 2) {
14         printf("1\n");
15         return 0;
16     }
17
18     // Dynamic Programming variables tracking the last 3 states
19     // w0 represents W(i-3), w1 represents W(i-2), w2 represents W(i-1)
20     long long w0 = 1; // W(0)
21     long long w1 = 1; // W(1)
22     long long w2 = 1; // W(2)
23     long long current_ways = 0;
24
25     // Compute the ways iteratively up to n
26     for (int i = 3; i <= n; i++) {

```

```
26     for (int i = 3; i <= n; i++) {
27         current_ways = w2 + w0; // W(i) = W(i-1) + W(i-3)
28
29         // Shift states forward for the next iteration
30         w0 = w1;
31         w1 = w2;
32         w2 = current_ways;
33     }
34
35     // Print the final result followed by a newline
36     printf("%lld\n", current_ways);
37
38     return 0;
39 }
40
```

	Input	Expected	Got	
✓	6	6	6	✓
✓	25	8641	8641	✓
✓	100	24382819596721629	24382819596721629	✓

Passed all tests! ✓

Correct

Marks for this submission: 10.00/10.00.

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